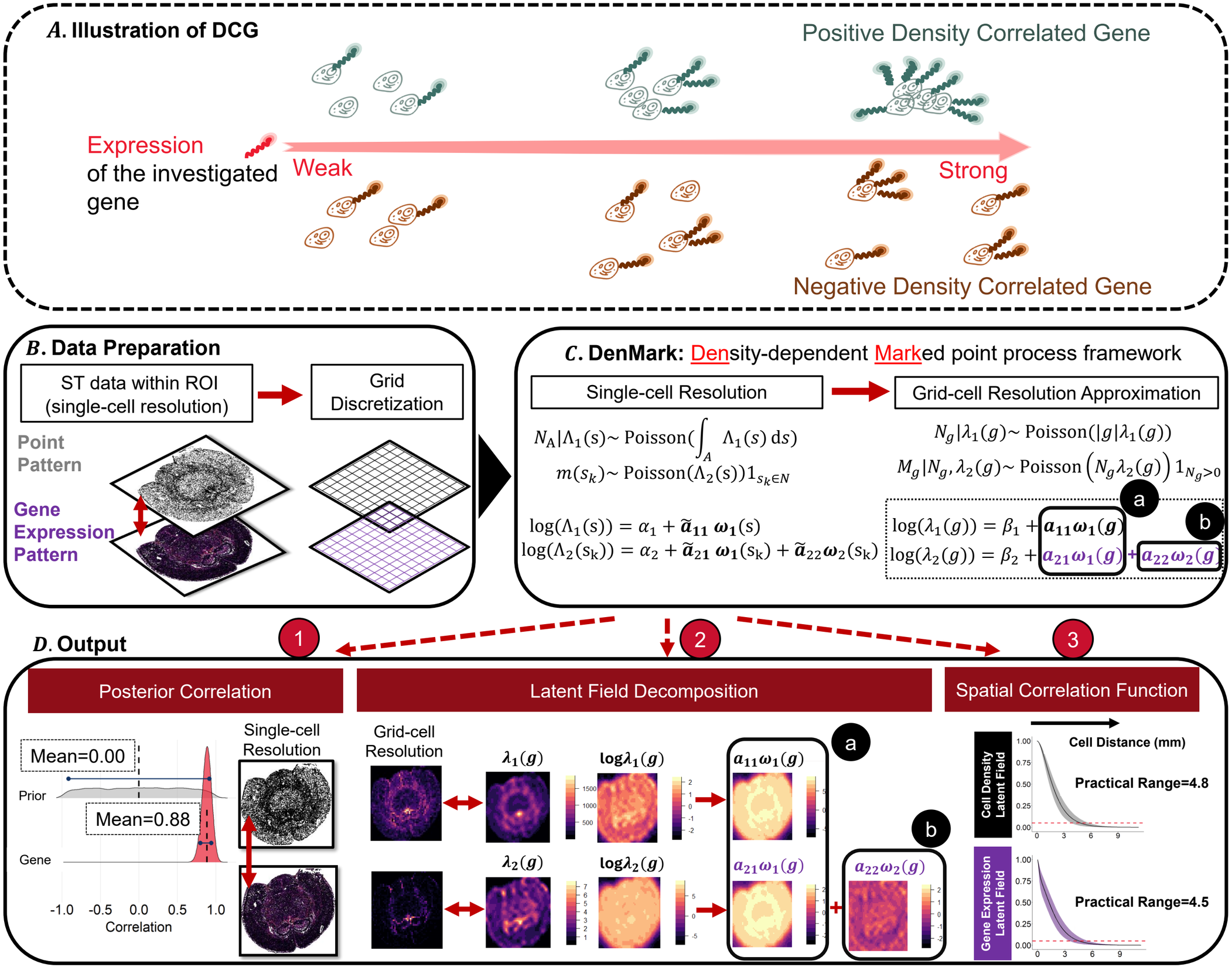
A. Illustration of DCG
Positive Density Correlated Gene
Expression of the investigated gene
Weak
Strong
Negative Density Correlated Gene
B. Data Preparation
ST data within ROI (single-cell resolution)
Grid Discretization
Point Pattern
Gene Expression Pattern
C. DenMark: Density-dependent Marked point process framework
Single-cell Resolution
Grid-cell Resolution Approximation
$N_A|\Lambda_1(s) \sim \text{Poisson}(\int_A \Lambda_1(s)\,ds)$
$m(s_k) \sim \text{Poisson}(\Lambda_2(s))1_{s_k \in N}$
$\log(\Lambda_1(s)) = \alpha_1 + \tilde{a}_{11}\,\omega_1(s)$
$\log(\Lambda_2(s_k)) = \alpha_2 + \tilde{a}_{21}\,\omega_1(s_k) + \tilde{a}_{22}\omega_2(s_k)$
$N_g|\lambda_1(g) \sim \text{Poisson}(|g|\lambda_1(g))$
$M_g|N_g, \lambda_2(g) \sim \text{Poisson}\left(N_g\lambda_2(g)\right)1_{N_g>0}$
$\log(\lambda_1(g)) = \beta_1 + a_{11}\omega_1(g)$
$\log(\lambda_2(g)) = \beta_2 + a_{21}\omega_1(g) + a_{22}\omega_2(g)$
a
b
D. Output
1
2
3
Posterior Correlation
Latent Field Decomposition
Spatial Correlation Function
Mean=0.00
Prior
Mean=0.88
Gene
-1.0 -0.5 0.0 0.5 1.0
Correlation
Single-cell Resolution
Grid-cell Resolution
$\lambda_1(g)$
$\log\lambda_1(g)$
$\lambda_2(g)$
$\log\lambda_2(g)$
$a_{11}\omega_1(g)$
$a_{21}\omega_1(g)$
$a_{22}\omega_2(g)$
Cell Distance (mm)
Cell Density Latent Field
Practical Range=4.8
Gene Expression Latent Field
Practical Range=4.5